

MediaOps CoPilot

Autonomous incident response for live video

When something breaks behind a live stream, every second matters. It detects, diagnoses, fixes, and verifies recovery — autonomously, before viewers notice.

This capability can be reused and scaled across the media & entertainment industry.

Autonomous Agentic AI Solution

Built on: Gemini · Google Cloud · Grafana · Firestore · Vertex AI · FFmpeg pipeline

Observability: Grafana + Prometheus

The problem

Detecting a stream failure is easy. Understanding and fixing it — especially during a live incident — is slow and costly

What used to happen before

- An alert fires — but only says something is wrong.
- An on-call engineer scrambles to investigate.
- Diagnose → decide → fix → check by hand.
- 15+ minutes pass while viewers see broken video and drop off.

After With MediaOps CoPilot

- The problem is detected and confirmed automatically.
- Multiple AI agents diagnose it from video and telemetry independently.
- A safe fix is authorized, executed, and verified.
- Resolved autonomously — in 30 to 90s, before viewers notice.

Who this is for: **live streaming services** — and broadcasters, OTT platforms, and live-event providers. The on-call engineers and ops teams who own uptime and carry the cost of every incident.

Value & go-to-market

Monitoring tells you something's wrong. **MediaOps CoPilot fixes it** — autonomously, in under 90 seconds.

Who it's for

- Live streaming services (OTT / platforms)
- Live sports streaming
- Broadcasters moving to cloud/IP
- Live-event providers

Where every second of downtime means lost viewers and revenue.

Adoption — a trust ramp

Observe → Assisted → Verified → Autonomous

Earn trust before handing over autonomous action.

The value we create



Recovery in ~1 min

vs. 15+ mins manual



Reduced outages

caught before viewers notice minimizes viewer impact, downtime, and business loss.



Engineer time back

no more repetitive 3am firefighting

Pricing

Tiered subscription by streams protected — self-serve for smaller operators, enterprise contracts for major platforms.

Priced as a fraction of the downtime and engineering costs we eliminate.

Built on Google Cloud + Grafana

Every integration is used at runtime — not just referenced.



Gemini 2.5 flash + Google ADK

Vision, Infra, and Remediation agents — reasoning at runtime via Vertex AI.



Vertex AI embeddings

Semantic retrieval of similar past incidents (RAG).



Firestore

Durable incident state, audit trail, and operational memory.



Grafana + Prometheus

Read-only telemetry via Grafana MCP; live dashboards.



Cloud Run

Deployed, public, always-on hosted service.



ffmpeg pipeline

Real video workload with genuine, recoverable faults.

Source code: github.com/charvimaurya/MediaOps

Model selection: why Gemini 2.5 Flash

A deliberate trade-off, not a default choice — weighed against Gemini Pro and Flash-Lite.

Why Gemini 2.5 Flash fits this project



Low latency

Important for fast incident response.



Lower cost / quota

Cheaper than larger Pro models to run continuously.



Strong multimodal support

Handles the Vision Agent's frame analysis.



Reliable classification

Consistent structured output for Infra and Remediation agents.



Vertex AI integration

Production availability, ready to deploy.

Alternatives considered



Gemini Pro — not selected

Could improve difficult reasoning, but adds latency and cost with little benefit for this project's narrow, well-structured tasks.



Flash-Lite — not selected

Cheaper to run, but risks reduced vision and classification reliability for the Vision and Infra agents.

The model supplies judgment only — symptom detection, fault classification, and remediation proposals. Deterministic components still control aggregation, safety approval, execution, and recovery verification.

What it does

A full autonomous incident lifecycle — no human in the loop for routine failures.

1

Detect

Deterministic monitor watches telemetry continuously.

2

Diagnose

AI reads the video symptom and the infrastructure metrics.

3

Decide

Proposes a fix from a fixed set of tested actions.

4

Gate

Deterministic safety check authorizes or blocks it.

5

Execute

Runs the approved fix against the real infrastructure.

6

Verify

Independently confirms recovery in both domains.

Core principle: AI supplies judgment · deterministic code verifies supplies authority

System architecture

Continuous monitoring → two AI agents investigate → confirm → safely fix → verify.

Always-on monitoring

A deterministic monitor watches the live metrics continuously and raises one incident per real problem.

AI investigation

Two Gemini agents investigate in parallel — Vision on the video, Infra on the telemetry — and their findings must corroborate.

Remediation Agent

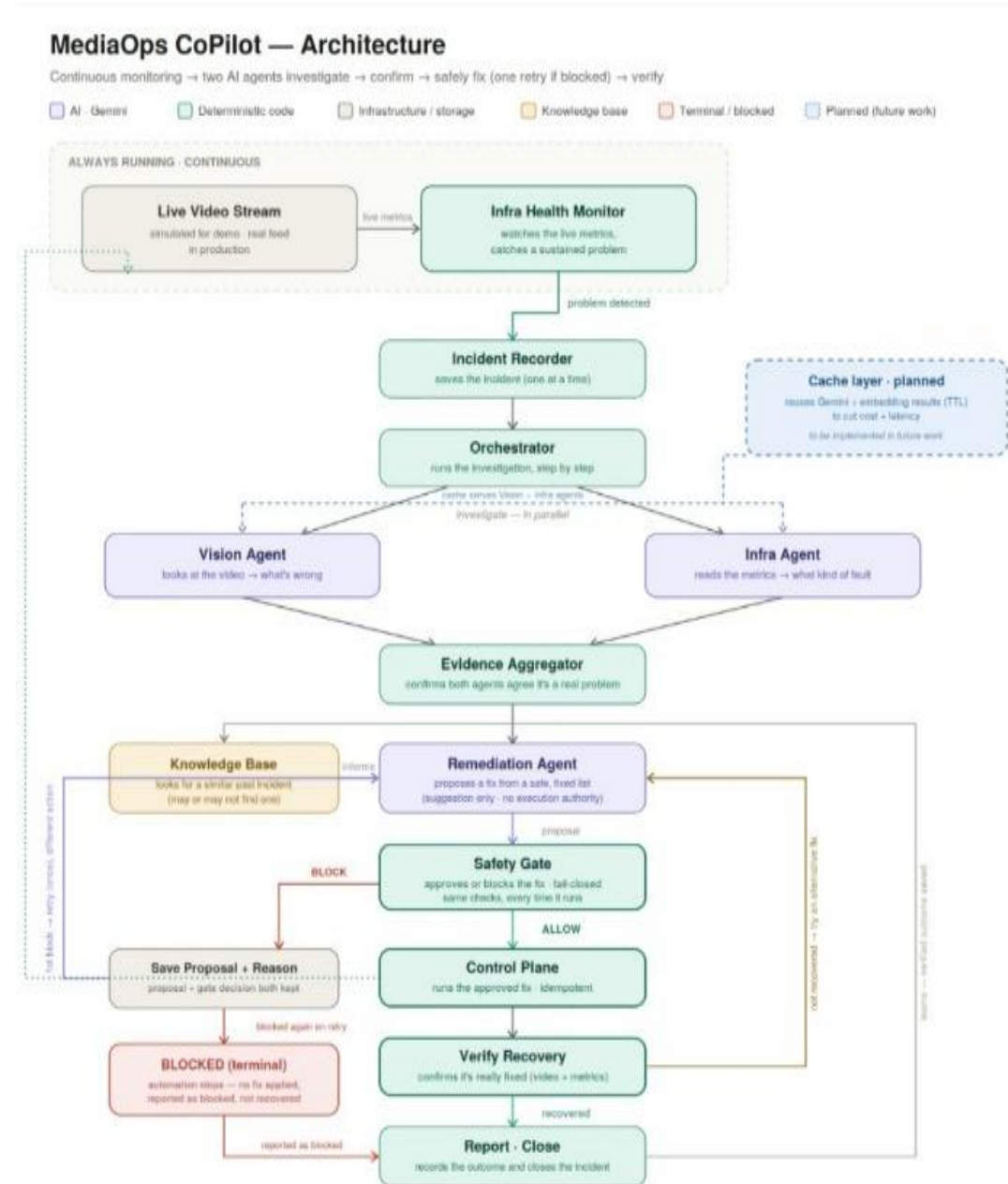
Weighs the evidence and past incidents and proposes one fix — from a fixed, safe set of actions. It recommends only; it never executes.

Guarded authority

A fail-closed Safety Gate authorizes the fix; the Control Plane executes it idempotently; recovery is independently verified.

Operational memory

Every verified recovery is saved back — the system learns from real incidents over time.



Safe by design – Guardrails

Autonomy is bounded by deterministic guardrails at every step.



Detection

Sustained-anomaly threshold — never fires on a transient blip.



Corroboration

Video and telemetry evidence must independently agree.



Fixed action set

AI can only choose from tested actions — it cannot invent commands.



Safety Gate

Deterministic, fail-closed — any error or bad state means BLOCK.



Idempotent execution

Duplicate delivery can never execute a fix twice.



Independent verification

"It ran" is not "it recovered" — recovery is proven, sustained.

The three AI agents

Each has one clear job — they investigate and recommend, but never act on their own.

1

Vision Agent

Looks at the actual video

Gemini reads a real frame from the stream and identifies the viewer-visible symptom — blocky, frozen, black, or fine. This is the "what does the viewer see?" evidence.

2

Infra Agent

Reads the infrastructure

Gemini interprets the live telemetry (CPU, frame rate, packet loss) via Grafana and classifies what kind of fault it is. This is the "what does the system say?" evidence.

3

Remediation Agent

Proposes the fix

Given both agents' evidence and similar past incidents, Gemini recommends one fix — chosen only from a fixed, tested set of actions. It proposes; it never executes.

Their findings are cross-checked: only when the Vision and Infra evidence agree is it confirmed a real incident.

Tested and guarded at every step

The AI is bounded by deterministic guardrails — and every component is tested against known faults.

Detection

Only a sustained problem raises an incident — never a one-off blip.

Evidence must agree

Video and telemetry are cross-checked; contradictory evidence is rejected.

Fixed action set

The AI can only choose tested actions — it cannot invent a command.

Safety Gate

A deterministic, fail-closed check approves or blocks every fix. Any error means block.

Safe execution

The approved fix runs once and only once — no duplicate actions.

Independent verification

Recovery must be proven in both video and metrics — "it ran" is not "it's fixed."

Every component is validated against a golden set of known faults, run repeatedly — and the AI agents reach 98% accuracy, with uncertain results caught by the guardrails rather than acted on.

The impact

~48s

average time to recovery, fully autonomous

15+ min

typical manual incident response

- Fewer viewer-impacting outages
- Engineers freed from repetitive incident detective work
- Consistent, auditable handling that improves over time

When something goes wrong behind the
stream every second matters and,
the viewer shouldn't have to know.

MediaOps CoPilot — an autonomous first responder for media infrastructure.